

Surgery for kidney cancer

Surgery can be used to treat kidney cancer. It will depend on the stage of your cancer and your general health.

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About surgery for kidney cancer

Surgery may be the only treatment you need. The type of surgery (operation) you have will depend on the [stage of the cancer](#) and your general health.

It is important to talk to your surgeon and specialist nurse about the operation before you have it.

Removing part of the kidney (partial nephrectomy)

This operation is called a partial nephrectomy or nephron-sparing surgery. It is the most common operation for kidney cancers smaller than 4cm. You may also have it for larger kidney cancers. Your doctor will explain if it is right for you.

During the operation, the surgeon removes the part of the kidney with the cancer. The aim is to remove the cancer but to leave as much normal tissue as possible. This is so the kidney can keep working.

You may have this operation to try and keep the affected kidney. This may be because:

- your other kidney does not work well
- you have kidney disease or other conditions that may affect how your kidneys work in the future.

If you have a type of kidney cancer that has been passed to you from a parent (inherited), you may be offered this operation. This is because there is a risk you may develop a cancer in the other kidney.

You are likely to be referred to a specialist hospital for this type of surgery. This means you may have to travel for treatment.

Removing the whole kidney (radical nephrectomy)

This operation is called a radical nephrectomy. You may have this operation if it is not possible to take only part of the kidney. For example, if the cancer is:

- affecting a large part of the kidney
- in the middle of the kidney.

During the operation, the surgeon removes the whole kidney and some surrounding tissue. They also remove [lymph nodes](#) close to the kidney to check them for cancer cells.

Removing a kidney is a big operation. It is not suitable for everyone. Your doctors may need to do tests to check you are well enough for surgery.

If there is cancer in 1 of the large blood vessels leaving the kidney, it may be possible to have surgery. But this is a more complicated operation. You will need to go to a specialist hospital for this surgery.

Can I live with one kidney?

You can usually live a normal life with 1 kidney. It can do the work of both kidneys. The surgeon can talk to you about this before you decide whether to have surgery.

We have more information about [making treatment decisions](#).

Removing the kidney when the cancer has spread

Your doctors may talk to you about an operation to remove the kidney even if the cancer has spread. It will not cure the cancer, but in some people, it may help them live longer. Your treatment plan will usually include other treatments such as [immunotherapy and targeted therapy](#).

You may need to think about the possible advantages of having the kidney removed, as well as the effects of having a big operation. Your doctors will talk to you about the possible benefits and disadvantages.

Surgery to remove a secondary cancer

Rarely, you may have an operation to remove a [secondary cancer](#) in another part of the body. This is a very specialised operation. It is usually only possible if the cancer has spread to 1 place in the body, such as the lung or brain. It is also important that you are well enough for the operation.

Removing a secondary cancer will not usually cure the cancer, but it may help control the cancer for longer. Or it may help reduce any symptoms it might be causing. Occasionally, it may be used to try to cure the cancer. Your cancer team can explain if removing the secondary cancer may help you.

Types of surgery

There are different ways a surgeon can operate on the kidney. Your surgeon will explain how they will do your operation and answer any questions you have. You usually have a general anaesthetic for kidney cancer surgery.

Kidney cancer surgery is often done as keyhole surgery, but open surgery is sometimes needed.

Keyhole (laparoscopic) surgery

During keyhole surgery, the surgeon uses a laparoscope to see and work inside the tummy (abdomen). A laparoscope is a thin tube with a light and camera on the end. It sends video images to a monitor.

The surgeon starts by making 3 or 4 small cuts in the tummy. The laparoscope then goes through 1 of the small cuts in the skin. The tummy will be inflated with gas to help the surgeon operate more easily.

During the operation, the surgeon puts the kidney they want to remove into a bag inside the body. At the end of the operation, they will remove the bag with the kidney inside by either:

- making 1 of the cuts bigger
- adding an extra cut.

The position of the cuts depends on who is having the surgery and how the surgeon does the operation. Your surgeon can talk to you about what to expect. Recovery from keyhole surgery is usually quicker than from open surgery. This is because the wounds are smaller.

Image: The illustration shows an example of where scars may be after an operation to remove the left kidney using keyhole surgery. This type of surgery is also sometimes called laparoscopic surgery. There are four small scars on the left side of the body. The largest scar is about 2 to 3cm in size. It begins just above the belly button, curves around to the left of it and ends just below it. The other scars are about 1cm in size. One is at about the same height as the belly button and the other two are higher up, one just above and one just below the waist.

Robot-assisted keyhole surgery

The surgeon may use a specialised machine to help with keyhole surgery. The surgeon controls the instruments that are attached to the machine. This is called robot-assisted surgery. The robot makes it possible for the surgeon to move in a very precise way to do complicated keyhole surgery.

Open surgery

During open surgery, the surgeon makes 1 large cut just under the bottom rib or between the 2 bottom ribs. This cut can be up to 30cm long. After the operation, the surgeon stitches or staples the wound closed.

Image: The illustration is a side view of a person's body. It shows the size and position of the scar left by open surgery to remove the left kidney. The scar begins at the back about the level of the lower rib and curves down towards the front. Alt text: The side-view illustration shows the scar's size and position from open surgery to remove the left kidney, starting at the back near the lower rib and curving down toward the front.

Before your operation for kidney cancer

Before your operation, you will visit a [pre-assessment clinic](#). You will be asked about your medical conditions and any medication you take.

You may have tests to check your general health. These can include:

- [blood tests](#)
- a chest [x-ray](#)
- a [recording of your heart \(ECG\)](#)
- tests to check your kidney function.

We have more information about [preparing for surgery](#).

Recovery after kidney cancer surgery

Your [recovery after surgery](#) will depend on the type of operation you have.

After the operation, you will be encouraged to start moving around as soon as possible. This can help reduce the risk of some problems.

Possible complications of surgery for kidney cancer

The most common complications after surgery are:

- a wound infection
- bleeding
- a chest infection
- a blood clot.

The nurses will monitor you for these. Tell them straight away if you:

- feel unwell
- have any bleeding
- notice swelling and redness in any of your arms or legs
- have symptoms of an infection – such as a cough, or leaking wound.

You will be given a telephone number to call if you need advice when you get home.

After open surgery to the kidney, some people develop a bulge (swelling) along their scar. Doctors call this an incisional hernia. It can happen because of weakness in the muscles around the scar. Tell your doctor or nurse if you develop a bulge around the scar.

We have more information about [general complications after surgery](#).

Going home after surgery

How long you stay in hospital depends on the type of operation you have and how quickly you recover.

After keyhole surgery, most people go home after 2 to 5 days. After open surgery, most people go home after 5 to 7 days.

Your body uses a lot of energy to heal. It is common to feel more tired than usual for about 6 weeks after the operation. Most people feel they have recovered after about 12 weeks.

You will not be able to do some everyday activities while you recover. For example, you may not be able to shop, drive, have sex, play sport or work. Your nurse or surgeon can tell you when you will be able to do these things again.

Before you leave hospital, you will be given an appointment for a check-up. This is usually about 6 weeks after your operation. It will be at an outpatient clinic. The appointment is a good time to ask questions, or talk about any problems you have after your operation.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our kidney cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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Date reviewed

Reviewed: 01 April 2025 | Next review: 01 April 2028

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help